A Phase-Ordered Pre-Geometric Projection Framework

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We propose an atemporal, pre-geometric substrate described purely by relational/algebraic structure. A compact internal SU(2)-like symmetry and a distinguished phase/action ordering generator are taken as primitive. A unique, necessary physical projection map yields an emergent manifold-like spatial structure (with effective dimension $D^* \approx 3$ in our regime), objective time-order (as a metric on phase order), and a finite distinguishability bound expressed in terms of a boundary cut-capacity functional (expected to reproduce an area-law in low-distortion manifold regimes). Quantum discreteness is treated as emergent from stable representation content under projection constraints, while gravitational geometry is defined as the output of the projection's correlation-based embedding and clock-rate reconstruction. In empirically accessed regimes the framework is required to recover standard General Relativity and quantum field theoretic predictions. Possible additional operational access to nonlocal substrate correlations (including any potential signaling via entanglement "junctions") is formulated as a constrained open module rather than assumed a priori.

I. REVIEWER-FACING SUMMARY

A. What this framework commits to

- **Substrate:** an atemporal, pre-geometric relational/algebraic structure $(\mathcal{A}, \omega)$ with a compact internal symmetry and a preferred **phase/action ordering flow** σ_s (order parameter s, not time).
- **Projection:** a **unique and necessary** physical map Π from substrate to effective descriptions. Π is defined operationally by (i) a stability-selected coarse-graining into finite-capacity "cells", (ii) correlation-defined locality, (iii) an embedding procedure into 3D, and (iv) a phase-order-to-clock-time mapping.
- **Finite distinguishability:** any finite projected region has bounded distinguishable information, scaling with **boundary area** (Planck-area-like scale ℓ_*^2), interpreted as a **relational encoding boundary** rather than a geometric boundary in the substrate.
- **Emergence:** discrete quantum "types" arise from stable representation content under projection constraints; effective geometry arises from correlation structure and the clock mapping, not from substrate curvature.

B. What this framework does *not* claim (yet)

- It does **not** assume a complete derivation of the Standard Model spectrum, coupling constants, or cosmological parameters in this draft.
- It does **not** assert superluminal signaling. "Junction access" is defined as a constrained open module with parameters required to match existing no-signaling constraints in all empirically accessed regimes.
- It does **not** claim a closed-form fundamental "constitutive law" $g = F(\rho, \ldots)$; the effective metric is defined by the explicit projection construction (graph distances $\rightarrow$ embedding $\rightarrow$ reconstructed h_{ab} and $d\tau$ mapping).

C. Scientific status and evaluation

- The framework is **empirically anchored** by matching requirements: in accessible regimes, the induced effective description must reproduce (to stated tolerance) standard GR tests and standard quantum statistics.
- The framework becomes **strictly falsifiable** when it asserts: (i) universal area-law capacity bounds outside their known domain, (ii) saturation/no-singularity behavior in regimes where GR predicts divergence, or (iii) any nonzero operational "junction access" parameter in regimes already constrained by experiment.
- Absent additional predictive commitments, the framework is evaluated by: (a) internal consistency (no hidden time/geometry in substrate), (b) uniqueness/minimality of Π under stated selection principles, and (c) whether GR/QFT emerge as stable effective closures with a small, auditable parameter budget.

II. MOTIVATION AND SCOPE

This document introduces a conceptual framework intended for comprehensive evaluation and iterative refinement. The goal is to provide a minimal set of primitives and postulates from which (i) an effective 3+1 spacetime description, (ii) quantum statistical behavior, and (iii) General Relativity in tested regimes can be recovered as projection-level physics. The framework is explicitly pre-geometric at the substrate level and treats time as emergent from a phase/action ordering structure.

Scope control: the present draft focuses on structural definitions, matching requirements, and evaluation criteria. It does not claim a completed derivation of the Standard Model or the Born rule; instead it identifies the minimal mathematical objects needed to attempt such derivations.

III. DESIGN CONSTRAINTS

- **No substrate time:** the substrate admits no fundamental temporal parameter.
- **No substrate spatial geometry:** the substrate is not a manifold with metric/curvature; geometric notions are projection outputs.
- **Analog intuition with bounded physics:** continuous symmetry/phase structure is allowed as an idealization, but no unbounded physical observables (no divergent densities; no infinite recursion depth).
- **Unique and necessary projection:** the mapping from substrate to projection is physical and not contingent among many alternatives.
- **Finite distinguishability:** any finite projected region has a bounded number of distinguishable states; capacity scales with boundary area (area law) via a relational encoding boundary.
- **Empirical recovery:** in currently tested regimes the framework must reproduce standard GR and quantum predictions to within experimental bounds.
- **FTL signaling remains an open module:** the existence of operational superluminal channels is not assumed; it is parameterized and constrained.

IV. CORE POSTULATES (AXIOMS)

A. Substrate postulates

S1 (Atemporal substrate). There exists a substrate $\mathcal{S}$ that is not spacetime and has no intrinsic time parameter.

S2 (Pre-geometric substrate). The substrate carries relational/algebraic structure but no spatial metric, curvature, or geometric topology in the usual sense.

S3 (Compact internal symmetry). The substrate admits a compact internal SU(2)-like symmetry acting on its relational degrees of freedom.

S4 (Phase/action ordering generator). A distinguished generator induces an ordering structure (phase/action order) that is not time but is used by projection to construct time-order.

B. Projection postulates

P1 (Equivalence-Class Uniqueness). There exists a physical projection map Π from substrate structure to projection-level effective physics. While the specific microscopic cell net may only be unique up to spontaneous symmetry breaking (attractor basin selection, analogous to a gauge choice), the *macroscopic predictions and geometric invariants* output by Π are uniquely determined, establishing equivalence-class uniqueness.

P2 (Emergent spatial structure; dimension as output). Π yields an emergent metric space with a manifold-like realization of some effective dimension D^*; in the empirically accessed regime of our universe, $D^* \approx 3$ (treated as an output/matching condition rather than an assumed input).

P3 (Emergent time-order). Π yields an objective time-order in projection as a metric on phase/action order; non-conscious systems inherit this time-order.

P4 (Finite distinguishability; boundary-capacity scaling). For any finite projected region R, physically distinguishable information is bounded by a boundary *cut-capacity* functional (defined on the correlation graph). In manifold-like regimes this scaling is expected to reproduce an area-law.

P5 (Bounded intensities). Projection-level densities and curvatures saturate rather than diverge; classical singularities indicate breakdown of the effective description.

C. Emergence postulates

E1 (Quantum discreteness from representations). Discrete quantum types arise from stable representation content of the internal symmetry under projection constraints, not from fundamental substrate discreteness.

E2 (Geometry as projection output). The effective metric $g_{\mu\nu}$ is constructed by the projection stages Π_{loc}, Π_{geom}, and Π_{time}: correlation-derived distances define an emergent locality graph; an embedding procedure reconstructs the spatial metric h_{ab}; and phase/action ordering sets the clock mapping via $d\tau = \beta_0 \cdot \exp(\Phi)\, dS_{\text{act}}$, where Φ is a Laplacian-derived potential on the correlation graph sourced by an entropy-contrast field $\delta\rho$. Encoding-density proxies are derived scalar summaries of local information/capacity structure and are not primitive geometric variables. Any apparent constitutive form

$g \approx F(\cdot)$ is understood as a phenomenological approximation to the reconstructed metric output by Π.

E4 (Lorentz Symmetry as IR Fixed Point). The algebraic commutation of the projection maps with the internal SU(2) symmetry ($E_i \circ \alpha_g = \alpha_g \circ E_i$) enforces a **Custodial Symmetry** on the emergent effective field theory. This symmetry suppresses the tree-level generation of dangerous dimension-4 Lorentz-violating operators that generically arise from lattice discreteness. We invoke a mechanism analogous to **spin-connection locking**. Because the internal SU(2) maps to the local frame fields (vielbeins) of the emergent tangent space, macroscopic spatial rotations SO(3) are dynamically identified with internal SU(2) gauge transformations. By demanding SU(2) equivariance, the coarse-grained nodes are forced to transform as isotropic SO(3) scalars. With these leading anisotropic artifacts eliminated, continuous Lorentz invariance becomes a highly stable **Infrared (IR) Fixed Point** of the Renormalization Group (RG) flow — analogous to the emergence of relativistic dispersion relations in discrete condensed-matter systems (e.g., graphene, Weyl semimetals). The framework therefore predicts that microscopic Lorentz violations persist at the extreme UV scale (of order ℓ_*), but are RG-irrelevant and suppressed by powers of $(\ell_*/L)^n$ at observable scales $L \gg \ell_*$. These residual violations constitute a falsifiable prediction.

J (module) (Junction accessibility). In all known regimes, operations available to agents obey no-signaling in local marginals. Additional junction access is treated as a constrained open module.

V. MATHEMATICAL PRIMITIVES

A. Substrate representation (analog-friendly)

Represent the substrate as a pair:

$$\mathcal{S} = (\mathcal{A}, \omega) \tag{1}$$

where $\mathcal{A}$ is an algebra of relational observables and ω is a state on $\mathcal{A}$. No manifold, metric, or time parameter is assumed at this level.

Analog-friendly choice. To support continuous internal symmetry/phase structure while preserving *finite physical realizability* (via projection-level finite distinguishability), take $\mathcal{A}$ to be a separable operator algebra that is well-approximated by finite matrix algebras at any finite resolution. A convenient template is:

$$\mathcal{A} = \mathcal{A}_{\mathrm{rel}} \otimes \mathcal{A}_F \tag{2}$$

where $\mathcal{A}_{\mathrm{rel}}$ is the "relational bulk" algebra (pre-geometric), chosen so that it can be approximated by an increasing sequence of finite-dimensional matrix algebras:

$$\mathcal{A}_{\mathrm{rel}} \approx \text{closure}\left(\bigcup_n M_{k_n}(\mathbb{C})\right) \tag{3}$$

This supports the **analog** intuition (continuous symmetry/phase) while remaining compatible with the principle that *only a finite number of states are physically distinguishable in any finite projected region* (Section 6).

$\mathcal{A}_F$ is an optional finite "internal" algebra used to encode gauge/representation structure in a purely algebraic way. A minimal candidate that naturally contains $U(1)$, $SU(2)$, and $SU(3)$-type unitary structure is:

$$\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}) \tag{4}$$

with $\mathbb{H}$ the quaternions.

Remark (no infinities physically instantiated). Continuous groups and infinite-dimensional algebras are treated as *idealized descriptions*: the framework's finiteness claim is that no *projection-level observable* diverges and that any finite region has finite operational capacity (area-law distinguishability), not that the descriptive mathematics cannot employ limits.

B. Internal symmetry action

Let α be an action of the internal symmetry group on $\mathcal{A}$ as *-automorphisms:

$$\alpha : \mathrm{SU}(2) \to \mathrm{Aut}(\mathcal{A}) \tag{5}$$

Symmetry invariants are quantities unchanged under α_U for all $U \in \mathrm{SU}(2)$.

C. Phase/action ordering generator

The substrate has no time, but it provides an **ordering structure** that projection uses to construct time-order.

(i) Generator form (working placeholder). Let $H \in \mathcal{A}$ be a distinguished self-adjoint element ($H = H^\dagger$) defining a one-parameter family of inner automorphisms:

$$\mathcal{A} \mapsto e^{isH} \mathcal{A} e^{-isH} \tag{6}$$

The parameter s is a phase/action **order parameter**, not substrate time.

(ii) Canonical-flow form (parameter-reducing option). Impose that the pair $(\mathcal{A}, \omega)$ determines a canonical one-parameter automorphism flow σ_s on $\mathcal{A}$. Projection-time is then constructed from monotone "distance" along this flow. In this view, H is not chosen freely; it is the generator associated with the canonical flow in a suitable representation.

D. Projection map (explicit construction template)

In this framework, the projection map Π is not treated as an arbitrary "fit function." Instead it is defined as a *constrained construction* that turns substrate relational structure into:

- an emergent 3D locality structure,
- an effective spatial metric,
- an objective time-order (from phase/action order),
- an encoding-density field and finite distinguishability bounds,
- effective subsystem states used for quantum predictions.

Mathematically, it is useful to write Π as a composition of four stages:

$$\Pi = \Pi_{\text{time}} \circ \Pi_{\text{geom}} \circ \Pi_{\text{loc}} \circ \Pi_{\text{res}} \tag{7}$$

1. *Stage 0: represent the substrate state (GNS representation)*

Given $(\mathcal{A}, \omega)$, choose the GNS triple $(\pi_\omega, \mathcal{H}_\omega, |\Omega\rangle)$ such that:

$$\omega(a) = \langle \Omega | \pi_\omega(a) | \Omega \rangle \quad \text{for all } a \in \mathcal{A}. \tag{8}$$

2. *Π_{res}: resolution-limited coarse-graining via split-property funnels*

A projection must provide a notion of subsystems/regions, but subsystems are not substrate primitives. Furthermore, strict local algebras in Quantum Field Theory are typically **Type III factors** (entanglement is ubiquitous), meaning they do not admit pure states or density matrices, and cannot be factorized into tensor products $\mathcal{A}_R \otimes \mathcal{A}_R^c$.

To resolve this **Type III incompatibility** without abandoning the continuum, we define emergent "cells" explicitly as **split-property funnels** (Type I approximants across algebraic buffer zones).

The Funnel Definition. Instead of a single algebra $\mathcal{A}_i$, the projection selects a **funnel structure** for each emergent site i:

$$\mathcal{A}_{i,\text{inner}} \subset N_i \subset \mathcal{A}_{i,\text{buffer}} \subset \mathcal{A} \tag{9}$$

where N_i is a **Type I factor** (isomorphic to $\mathcal{B}(\mathcal{H})$ or a finite matrix algebra $M_d(\mathbb{C})$ in toy models) that acts as a local "distinguishable semantic shell" between the inner core and the buffer zone.

The effective local algebra is identified with the Type I factor N_i. This guarantees the existence of well-defined local density matrices, entropies, and particle-like excitations.

Coarse-Graining. The coarse-graining maps E_i are conditional expectations onto these Type I factors:

$$\{\mathcal{A}_i \subset \pi_\omega(\mathcal{A})''\}_{i \in V}, \quad \text{with} \quad \mathcal{A}_i \cong M_{d_i}(\mathbb{C}), \tag{10}$$

together with conditional expectations (coarse-graining maps):

$$E_i : \pi_\omega(\mathcal{A})'' \to \mathcal{A}_i \tag{11}$$

that are completely positive, unital, and idempotent ($E_i \circ E_i = E_i$).

For any finite set of cells $R \subset V$, define the induced effective algebra and reduced state:

$$\mathcal{A}_R \approx \bigotimes_{i \in R} \mathcal{A}_i, \quad E_R := \bigotimes_{i \in R} E_i, \quad \omega_R := \omega \circ E_R. \tag{12}$$

Finite distinguishability constraint (non-geometric form). To avoid circularity, the capacity bound is imposed **before** any geometric embedding. Let $\mathrm{Cap}(\partial R)$ denote a **cut-capacity** functional on the cell net:

$$\mathrm{Cap}(\partial R) := \sum_{i \in R, j \notin R} \kappa(I_{ij}), \tag{13}$$

where I_{ij} is the mutual information between cells i and j, and κ is a fixed monotone map to dimensionless edge capacities.

Finite distinguishability constraint (Araki-relative form). For Type III compatibility, we eschew the naive von Neumann entropy. The bound is imposed on the **Araki relative entropy** of the region's state $\omega|_R$ against the canonical **KMS vacuum baseline** $\omega^{\text{vac}}|_R$:

$$S_{\text{Araki}}(\omega|_R \| \omega^{\text{vac}}|_R) \leq \eta \cdot \mathrm{Cap}(\partial R) \tag{14}$$

A geometric area scaling $\mathrm{Cap}(\partial R) \propto A_{\text{geo}}(\partial R)/\ell_*^2$ is treated as an emergent behavior in the low-stress embedding regime (§4.4.4), not as an input constraint.

Selection Principle E: symmetry-commuting, phase-flow stable cell net

The choice of the effective cell algebras $\{\mathcal{A}_i\}$ and coarse-graining maps $\{E_i\}$ is fixed by a *stability principle* tied to the phase/action ordering flow σ_s. Intuitively: the correct decomposition is the one for which phase/action ordering does not continually "slosh" information across emergent cell boundaries.

Let α_g denote the internal SU(2)-like symmetry action on $\mathcal{A}$. Let σ_s denote the phase/action ordering flow. Define $\mathfrak{E}_{\text{adm}}$ as the set of families $\{E_i\}$ (and their associated cell algebras $\{\mathcal{A}_i\}$) satisfying all of:

1) **Finite-capacity cells.** Each cell algebra is finite-dimensional $\mathcal{A}_i \cong M_{d_i}(\mathbb{C})$, with either a fixed common dimension $d_i = d$ for all cells, or an explicit local bound $d_i \leq d_{\max}$.

2) **Symmetry commutation (internal isotropy constraint).** Each coarse-graining map commutes with the SU(2)-like symmetry:

$$E_i \circ \alpha_g = \alpha_g \circ E_i \quad \text{for all } g \in \mathrm{SU}(2) \text{ and all } i. \tag{15}$$

The "Custodial Symmetry" Mechanism (Lorentz Protection). This constraint is the physical reason why the discrete graph does not break Lorentz symmetry at macroscopic scales. We invoke a mechanism analogous to **spin-connection locking**. Because the internal SU(2) maps to the local frame fields (vielbeins) of the emergent tangent space, macroscopic spatial rotations SO(3) are dynamically identified with internal SU(2) gauge transformations. By demanding SU(2) equivariance, the coarse-grained nodes are forced to transform as isotropic SO(3) scalars. This internal custodial symmetry strictly suppresses the generation of dangerous tree-level dimension-4 Lorentz-violating spatial lattice artifacts (e.g., $k^\mu k^\nu \Delta u_\mu u_\nu$). With these leading anisotropic artifacts eliminated, continuous Lorentz invariance becomes a highly stable **IR fixed point** of the RG flow.

3) **Information retention (anti-triviality constraint).** Impose a global bound on relative-entropy loss:

$$D(\omega \| \omega \circ E) \leq \varepsilon, \tag{16}$$

Primary stability functional (phase-flow leakage). For each admissible $\{E_i\}$, define the phase-flow leakage:

$$\mathcal{L}_{\mathrm{leak}}(E) := \int ds\, w(s) \sum_i \|E_i \circ \sigma_s - \sigma_s \circ E_i\|^2, \tag{17}$$

where $w(s) \geq 0$ is a weight over a chosen phase-order interval. Exact co-motion corresponds to $\mathcal{L}_{\mathrm{leak}} = 0$.

Optional tie-breaker (local information drift). Define $\omega_s := \omega \circ \sigma_s$, and let $\rho_i(s)$ be the density operator corresponding to the reduced state $\omega_s \circ E_i$. For a small step δ, define:

$$\mathcal{L}_{\mathrm{drift}}(E) := \int ds\, w(s) \sum_i \frac{1}{\delta^2} D(\rho_i(s+\delta) \| \rho_i(s)). \tag{18}$$

Cell selection as causal gradient flow. The cell net $E^* = \{E_i^*\}$ is **not** defined by a global argmin. Instead, it is defined as the **stable fixed-point (attractor)** of a local, causally-bounded gradient flow on the space of admissible coarse-grainings.

Local update channel. For each cell i, define the local leakage gradient:

$$\delta_i \mathcal{L}_{\mathrm{leak}} := \left. \frac{\partial \mathcal{L}_{\mathrm{leak}}}{\partial E_i} \right|_{E_{j\neq i} \text{ fixed}} \tag{19}$$

To strictly avoid circularity with the emergent geometry d_G, this gradient depends only on cells j within the **Algebraic Lieb-Robinson neighborhood** $N_{\mathrm{LR}}^{\mathrm{alg}}(i)$ of cell i. This neighborhood is defined intrinsically on the pre-geometric substrate using operator commutators under the canonical flow, completely independent of the correlation graph:

$$N_{\mathrm{LR}}^{\mathrm{alg}}(i) := \{ j \in V : \|[\sigma_{\Delta s}(\mathcal{A}_i), \mathcal{A}_j]\| > \epsilon \} \tag{20}$$

Causal flow. The coarse-graining maps evolve under a local descent:

$$\partial_s E_i = -\eta_{\mathrm{flow}} \cdot \Pi_{\mathrm{adm}} \left[\delta_i \mathcal{L}_{\mathrm{leak}} + \lambda_{\mathrm{drift}} \cdot \delta_i \mathcal{L}_{\mathrm{drift}} \right] \tag{21}$$

Causality guarantee. Because $\delta_i \mathcal{L}_{\mathrm{leak}}$ depends only on E_j for $j \in N_{\mathrm{LR}}^{\mathrm{alg}}(i)$, updates at cells i and k with non-overlapping algebraic neighborhoods **strictly commute**:

$$[\partial_s E_i, \partial_s E_k] = 0 \quad \text{whenever } N_{\mathrm{LR}}^{\mathrm{alg}}(i) \cap N_{\mathrm{LR}}^{\mathrm{alg}}(k) = \emptyset. \tag{22}$$

Definition of the selected cell net. The projection-level cell net is the fixed point of this flow:

$$E^* := \lim_{s\to\infty} E(s), \quad \text{where } \left. \partial_s E_i \right|_{E=E^*} = 0 \quad \text{for all } i. \tag{23}$$

Variational interpretation (Thermodynamic, not Computational). The causal gradient flow defined above is a *physical relaxation process*, not an algorithmic instruction. No agent or physical subsystem solves an NP-hard global optimization. The flow is analogous in spirit to the least-action principle: the cell net that "exists" at projection level is the one that the local thermodynamic relaxation has converged to. In toy-model simulations (where $N \leq 12$ qubits), the fixed point may be located by exhaustive enumeration as a computational shortcut; this does not alter the physical definition, which remains the causal flow attractor.

Crucially, this relaxation requires the emergent subsystems to act as **effectively open thermodynamic systems**. The "environment" here is not an external physical universe, but rather the highly entangled, non-local UV degrees of freedom (the deep Type III commutant and algebraic buffer zones) explicitly traced out by the Type I funnels. The macroscopic geometric cells survive by shedding interaction-induced phase-flow entropy into this unobservable, short-scale quantum vacuum. Without this mechanism, decoherence universally shatters the geometric condition number $\kappa(M) \to \infty$, triggering spatial collapse.

3. Π_{loc}: *locality from correlations (graph metric from mutual information)*

Define the mutual information between cells i and j via **Araki relative entropy**:

$$I_{ij} = S_{\mathrm{Araki}}(\omega_{i\cup j} \| \omega_i \otimes \omega_j) \tag{24}$$

Distance kernel. Define an emergent pairwise distance by a fixed monotone decreasing map f:

$$d_{ij} = \ell_* \cdot f(I_{ij}/I_0), \tag{25}$$

Markovian Geometric Filtering & Weighted Graph. Raw mutual information can falsely identify highly entangled but geometrically distant nodes (e.g., Bell pairs, error-correcting codes) as "adjacent." To ensure strict geometric proximity, the edge weight kernel applies a **Quantum Conditional Mutual Information (QCMI)** filter. If the correlations between i and j are entirely mediated by an intermediate neighborhood k, their QCMI vanishes: $I(i : j|k) \approx 0$. By restricting continuous edges to pairs with strictly non-Markovian (direct) QCMI, the framework systematically separates true geometric proximity from long-range topological entanglement. The final weighted graph is defined as $w_{ij} := \kappa(I_{ij})$ applied only to unscreened pairs.

Graph geodesic distance. With edge lengths d_{ij}, define graph-geodesic distance:

$$d_G(i,j) = \inf_{\text{paths } i\to j} \sum_{(a,b)\in\text{path}} d_{ab}. \tag{26}$$

4. Π_{geom}: *emergent geometry via complexity-stress minimization*

Ontic vs. Epistemic distinction. The physical (ontic) reality in this framework is solely the **discrete correlation graph** (V, d_G). The embedding into $\mathbb{R}^{D^*}$ described below is strictly an **epistemic coordinate chart**: a lossy compression constructed by macroscopic observers to extract the effective dimension D^*.

1. Dimension Selection (Derived, not assumed). The emergent dimension D^* is the integer that minimizes a **Complexity-Stress Functional** $F(D)$ that balances embedding fidelity against topological complexity:

$$D^* := \operatorname*{argmin}_{D\in\mathbb{N}} \left[\text{Stress}(D) + \lambda_{\text{dim}}|D - D_S|^2\right] \tag{27}$$

where Stress(D) is the standard MDS stress, D_S is the **Spectral Dimension** of the graph, and λ_{dim} is a penalty weight ("topological inertia"). The penalty weight λ_{dim} is evaluated in the degeneracy-breaking limit ($\lambda_{\text{dim}} \to 0^+$). It is not a tunable control parameter to force $D = 3$; it acts purely as a physical "topological inertia" to break ties between equally low-stress embeddings, preventing high-frequency quantum noise from causing the macroscopic dimensionality to jitter.

2. Relational Embedding (Epistemic Coordinates). Given D^*, the coordinates $\{x_i\}$ are determined by the stress-minimizing configuration in $\mathbb{R}^{D^*}$.

Local flatness as the Einstein Equivalence Principle. MDS inherently seeks the lowest-stress Euclidean embedding, which means it tends to "flatten" local neighborhoods into Euclidean patches. This is the framework's **native realization of the Einstein Equivalence Principle (EEP)**.

Curvature bypasses the flat coordinates. The true non-linear curvature of the emergent geometry is **not** computed from the MDS embedding coordinates. Instead, it is computed intrinsically on the raw, unflattened graph distances via Discrete Regge Calculus.

3. Local Metric Reconstruction with Relational SPD Regularization.

$$\min_{h\succ 0} \sum_{j\in N(i)} \left(d_G(i,j)^2 - \Delta x^T h \Delta x\right)^2 + \lambda_{\text{spd}}\|h - h_{\text{cov}}^{-1}\|_F^2 \tag{28}$$

4. Definition of Geometric Singularities (Horizons). A **Geometric Horizon/Singularity** is defined as a region where the condition number κ of the design matrix M for the metric fit diverges: $\kappa(M) \to \infty$.

5. Π_{time}: *unified emergent time via non-local graph potential*

1. The Clock-Rate Potential Equation.

$$(\Delta_w + \mu^2 I)\Phi = \delta\rho \tag{29}$$

where:

- Δ_w is the graph Laplacian on the mutual-information weighted graph: $(\Delta_w\Phi)_i := \sum_j w_{ij}(\Phi_i - \Phi_j)$.
- $\delta\rho$ is the source term (defined below).
- μ is a screening mass. In the continuum limit, setting $\mu = 0$ (massless) enforces the standard GR $1/r$ long-range behavior. On **finite discrete graphs**, however, the graph Laplacian Δ_w possesses a non-trivial kernel (constant mode), making the system singular at $\mu = 0$. Choosing a small $\mu > 0$ acts as a symmetry-preserving infrared (IR) regulator.

Note: This global Laplacian acts purely as an **elliptic constraint equation** on the foliation (mathematically identical in spirit to the Hamiltonian constraint in the ADM formulation of GR), not as an acausal dynamical signal propagator.

2. The Source Term: Temporal-Filtered Araki Contrast.

$$\delta\rho_i := -\frac{1}{s_0}\left[\bar{S}_{\text{Araki}}(\omega_i\|\omega_i^{\text{vac}})\right] \tag{30}$$

Physical interpretation: Relative Entropy Duality. Any physical excitation, regardless of its internal structure, yields a positive relative entropy and therefore a **negative source** $\delta\rho_i < 0$. This creates a universal duality between highly ordered matter (particles) and maximally mixed concentrations (black holes). Because $\delta\rho < 0$ **universally**, the screened Poisson equation natively generates **attractive gravity wells** ($\Phi < 0$). Clocks slow down, natively reproducing gravitational redshift, Shapiro delay, and the Newtonian $\Phi \sim -GM/r$ potential.

3. The Baseline Vacuum. ω^{vac} is defined as the **KMS (Kubo-Martin-Schwinger) thermal state**.

4. Emergent Proper Time. The local proper time τ along a phase trajectory is then constructed by integrating the **Laplacian-derived clock rate**:

$$d\tau(i) = \beta_0 \cdot \exp(\Phi_i)\, dS_{\text{act}} \tag{31}$$

where dS_{act} is operationally defined as the incremental trace-distance advanced by the canonical flow σ_s on the substrate algebra.

Minimal emergent spacetime line element (preferred slicing).

$$ds^2 = -c^2 d\tau^2 + h_{ab}(x)dx^a dx^b, \tag{32}$$

E. Symbol glossary

TABLE I. Symbol glossary

Symbol	Meaning
$\mathcal{S}$	Substrate object
$\mathcal{A}$	Substrate algebra of relational observables
$\mathcal{A}_{\text{rel}}$	Relational "bulk" algebra (pre-geometric)
$\mathcal{A}_F$	Finite internal algebra encoding gauge/representation structure
ω	State on $\mathcal{A}$
α	Internal symmetry action on $\mathcal{A}$
H	Phase/action ordering generator (not substrate time)
Π	Unique physical projection map
M^3	Emergent 3D spatial structure
$g_{\mu\nu}$	Emergent spacetime metric in projection
t	Emergent time coordinate/order metric in projection
$\rho(x)$	Encoding density field in projection
ℓ_*	Minimal encoding resolution scale (Planck-like)
η	Dimensionless coefficient in the area-law capacity bound
γ	Junction-access gating parameter (open module)
Δ	Junction-access correction functional (open module)

F. Substrate algebra and parameter budget

Substrate data. The substrate is specified by: $\mathcal{S} = (\mathcal{A}, \omega, \alpha)$. Continuous symmetry and continuous phase/action order are treated as idealized descriptions. Physical finiteness is enforced at the projection level via the area-law bound and bounded intensities.

G. Parameter budget (what must be fixed or fitted)

Structural (discrete) choices: (Architecture decisions) Choice of algebra class for $\mathcal{A}_{\text{rel}}$; choice of internal algebra $\mathcal{A}_F$; compact internal symmetry group; equivalence-class uniqueness of projection postulate. *Must be defended by conceptual minimality and consistency.*

Empirical scale-setting constants: (Expected to be measured) ℓ_* (minimal encoding patch); η (area-law coefficient); effective constants matching $(c, \hbar, G, \Lambda)$.

Projection response functions: (Primary risk of "parameter fitting") Mapping from encoding density to metric scaling; mapping from phase-order to clock time. *Must be restricted to minimal families with clear constraints and matching requirements.*

Substrate state selection: Choice/characterization of ω. *Must be constrained by a selection principle (symmetry, equilibrium relative to canonical flow, minimal information, etc.) to avoid embedding arbitrary structure by hand.*

Open junction module: γ(regime) and $\Delta(\ldots)$. *Must satisfy $\gamma \approx 0$ in all tested regimes; left open otherwise, but constrained by normalization/positivity and by compatibility with known no-signaling bounds.*

VI. PROJECTION OUTPUTS AND EMERGENT TIME

Time does not exist at the substrate level. Instead, projection constructs an objective time-order by mapping phase/action order to a temporal metric: $t = f(\Phi)$ with $f' > 0$.

VII. FINITE DISTINGUISHABILITY AND PLANCK-SCALE RESOLUTION

1. Algebraic Cut-Capacity. Let $R \subset V$ be a finite set of emergent funnels. Define the **cut-capacity** $\text{Cap}(\partial R)$ purely from the correlation graph weights:

$$\text{Cap}(\partial R) := \sum_{i \in R, j \notin R} \kappa(I_{ij}) \tag{33}$$

2. The Araki Entropy Bound. The distinguishable information content of R is limited by the cut-capacity:

$$S_{\text{Araki}}(\omega|_R \| \omega_{\text{vac}}|_R) \leq \eta \cdot \text{Cap}(\partial R) \tag{34}$$

3. Theorem of Emergence (Not an Axiom). If the correlation graph admits a low-distortion embedding into $\mathbb{R}^3$, then: $\text{Cap}(\partial R) \propto A_{\text{geo}}(\partial R)/\ell_*^2$.

The operational consequence of the capacity bound is a limit on the effective dimension of the Type I funnel factor N_R:

$$\dim(N_R) \lesssim \exp\big(\eta \cdot \mathrm{Cap}(\partial R)\big) \tag{35}$$

VIII. QUANTUM STATISTICS AS PROJECTION-LIMITED INFERENCE

Measurement outcomes are computed by the Born-rule form in empirically accessed regimes: $P(i) = \mathrm{Tr}(\rho_R E_i)$.

IX. EMERGENT GEOMETRY AND GR MATCHING

A. Metric definition as projection output

The projection pipeline replaces the role of a constitutive law by constructively generating geometry from correlation.

Entanglement Shear: the kinematic bridge from Φ to h_{ab}. Because Φ governs the local rate of temporal evolution (clock dilation), adjacent nodes i, j sitting at different gravitational depths ($\Phi_i \neq \Phi_j$) evolve at **differential rates** under the phase-order flow σ_s. This actively **shears** the bipartite quantum state ω_{ij}, altering the mutual information I_{ij}. Since the spatial metric h_{ab} is reconstructed from these graph distances, any gradient in Φ continuously warps h_{ab} through the quantum substrate itself, providing a strictly **kinematic quantum mechanism** for the non-linear tensor curvature of General Relativity.

B. GR matching as a discrete closure condition

1. *Primary Definition: Discrete Regge Closure*

The comparison is performed natively on the discrete pre-geometric structure.

1. **Intrinsic Discrete Geometry:** To ensure curvature remains strictly intrinsic and independent of the epistemic MDS embedding, we abandon coordinate space entirely. We construct a **Vietoris-Rips simplicial complex** $\mathcal{T}_{\mathrm{VR}}$ directly from the raw correlation metric space (V, d_G). This mathematically guarantees that the emergent topology and the subsequent Regge deficit angles are derived purely from relational quantum correlations, without any "Euclidean smuggling" from flat coordinate charts. *(Note: In finite low-distortion toy models, a Delaunay triangulation on the MDS embedded points is frequently utilized as a computationally efficient, homeomorphic proxy for the exact intrinsic VR complex).*

2. **Discrete Metric:** Assign edge squared-lengths $l_{ij}^2 = h_{ab}(x_i)\Delta x^a \Delta x^b$ consistent with the reconstructed local metric.

3. **Discrete Curvature (Regge):** Compute the curvature using **Regge Calculus**. The curvature is concentrated on the $(D^* - 2)$-dimensional bones (hinges). For $D^* = 3$, these are edges. The deficit angle ϵ_h at hinge h is:

$$\epsilon_h = 2\pi - \sum_{\mathrm{cell} \supset h} \theta_{\mathrm{cell}}(h) \tag{36}$$

where $\theta_{\mathrm{cell}}(h)$ is the dihedral angle of the tetrahedron at hinge h.

2. *The Closure Mismatch Functional*

Define the **discrete Regge Einstein tensor** G_h on each hinge h:

$$G_h \cdot l_h := \epsilon_h - \Lambda V_h \tag{37}$$

Define the **closure mismatch** $M(L)$ over a region V_L:

$$M(L) := \big\| G_{\mathrm{Regge}} - 8\pi G\, \Pi_{\mathrm{graph}}(T^{\mathrm{eff}}_{\mu\nu}) \big\|_L \tag{38}$$

In this discrete mapping, the scalar Araki-contrast source $\delta\rho$ explicitly acts as the effective energy density component (T_{00}) driving the Lapse field Φ. The full effective stress-energy tensor $T^{\mathrm{eff}}_{\mu\nu}$ is matched dynamically via the spatial derivatives of the entanglement shear applied to the local metric h_{ab}.

Weak-field consistency via discrete diagnostics. We extract an effective mass density ρ_{mass} from the discrete potential using the **discrete Poisson equation**:

$$4\pi G \rho_i \approx (\Delta_{\mathrm{VR}}\Phi)_i \tag{39}$$

X. STANDARD MODEL COMPATIBILITY (PROGRAM)

A commonly studied finite algebra whose unitary structure can realize the Standard Model gauge group is: $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$. Treat each effective cell as carrying an internal factor: $\mathcal{A}_i \approx M_{d_i}(\mathbb{C}) \otimes \mathcal{A}_F$.

XI. ENTANGLEMENT AND JUNCTION ACCESSIBILITY (OPEN MODULE)

For accessible operations and regimes: $P(b|x,y) = P(b|x)$ (within experimental bounds). To keep the possibility of additional operational access open without asserting it, introduce a gating parameter $\gamma(\text{regime}) \in [0,1]$ and write:

$$P(b|x,y) = \mathrm{Tr}(\rho_B E_b^{(x)}) + \gamma \cdot \Delta_b(x, y; \rho_{AB}) \tag{40}$$

XII. EVALUATION, FALSIFICATION, AND TEST PROGRAM

A. Mathematical status (definition / theorem / conjecture / matching)

Substrate triple $(\mathcal{A}, \omega, \alpha)$**:** *(Definition)* The substrate is specified as an atemporal, pre-geometric relational algebra with a compact internal symmetry action. **Failure mode:** Pure definition; failure only if later steps implicitly add substrate spacetime structure.

Phase/action ordering flow σ_s**:** *(Definition / Conjecture)* A distinguished ordering flow exists; s is an order parameter, not time. **Failure mode:** If derived (canonical/modular flow): show existence conditions on $(\mathcal{A}, \omega)$. If postulated: keep minimal and symmetry-compatible.

Cell net $\{\mathcal{A}_i, E_i\}$**:** *(Definition)* Projection-resolution stage outputs effective "cells" (Type I approximants / toy matrix factors) and coarse-graining maps. **Failure mode:** Toy models: explicit construction. General AQFT: implement via split-property funnels; failure if no consistent finite-resolution approximation exists.

Symmetry-commutation: *(Constraint)* $E_i \circ \alpha_g = \alpha_g \circ E_i$ for all $g \in \mathrm{SU}(2)$. **Failure mode:** Check directly in constructions; failure if recovered physics requires symmetry-breaking at the projection-selection level.

Stability selection $\mathcal{L}_{\mathbf{leak}}$**:** *(Definition)* Among admissible coarse-grainings, select those minimizing phase-flow leakage across cell boundaries. **Failure mode:** Toy models: compute minimizers numerically; failure if the minimizer is always trivial (information-destroying) despite retention constraints.

Existence of minimizers: *(Theorem / Conjecture)* Minimizers exist for finite toy systems and well-posed admissible sets; general operator-algebraic existence is open. **Failure mode:** Toy models: compactness/finite search; general case: requires technical assumptions on admissible channel set and topology.

Capacity bound $\mathrm{Cap}(\partial R)$**:** *(Definition)* Finite distinguishability is bounded by a non-geometric boundary-capacity functional computed from correlation weights. **Failure mode:** Pure definition; failure only if it cannot be made compatible with recovered continuum behavior.

Emergent "area law" scaling: *(Matching output)* In manifold-like regimes, $\mathrm{Cap}(\partial R) \propto A_{\mathrm{geo}}(\partial R)/\ell_*^2$. **Failure mode:** Validate in toy models that yield low-stress embeddings; falsified if scaling is generically volume-like even in 3D-like regimes.

Emergent locality I_{ij}, d_G**:** *(Definition)* Locality is defined from QCMI-screened correlations and multi-hop routing on weighted graphs. **Failure mode:** Definition is checkable; failure mode is pathological graphs (disconnectedness, non-manifold structure) for physically relevant states.

Dimension D^***:** *(Theorem + Matching)* D^* is selected by a distortion/complexity criterion. **Validated:** correctly recovers $D^* = 1$ for exact 1D Heisenberg chain and $D^* = 2$ for exact 2D Heisenberg grid. **Failure mode:** Falsified if D^* is unstable or consistently far from the intrinsic dimension in GR/QFT-like toy states.

Spatial metric $h_{ab}(x)$**:** *(Definition)* The emergent spatial metric is reconstructed from neighbor distances with explicit positive-definiteness constraints. **Failure mode:** Validate numerically; failure if SPD enforcement destroys the ability to match observables.

Clock-rate field β **and time** τ**:** *(Definition)* Emergent time is defined as a conversion from phase-order length to clock time, with β sourced by an entropy-contrast potential. **Failure mode:** Validate weak-field behaviors in worked examples; falsified if exterior/vacuum tails cannot reproduce redshift/Shapiro-like effects without tuning.

Spacetime metric $g_{\mu\nu}$**:** *(Definition)* g is defined by the embedding-derived h plus the clock mapping τ (no separate constitutive law $g = F(\rho)$). **Failure mode:** Pure definition; key is whether the reconstructed discrete geometry (Vietoris-Rips complex + Regge curvature) satisfies GR closure natively.

GR closure mismatch $M(L)$**:** *(Definition + Matching)* The discrete Regge Einstein tensor G_{Regge} on the simplicial complex must satisfy Einstein closure within tolerance. **Failure mode:** Falsified if no small-parameter instantiation yields small $M(L)$ across standard tests.

Born-rule statistics: *(Matching)* Outcome statistics must match standard QM in all currently tested regimes. **Failure mode:** Falsified by any predicted deviation already excluded; derivation from projection-limited inference is an open program item.

Operational no-signaling: *(Matching)* Local marginals must not depend on distant choices (junction gate $\gamma \approx 0$). **Failure mode:** Falsified by any proposal that enables controllable signaling in ordinary Bell-test regimes.

Junction-access deviations: *(Open module)* Whether additional operational access exists in extreme regimes is left open and parameterized. **Failure mode:** Must remain consistent with existing constraints; becomes testable if a concrete activation regime is specified.

Local Lorentz covariance: *(Matching)* Despite a preferred phase-order foliation in the construction, local Lorentz symmetry must emerge as an excellent IR symmetry via spin-connection locking. **Failure mode:** Constrained by precision Lorentz-violation bounds; falsified if implied violations exceed limits.

B. Internal validation tests (toy-model / simulation tests)

T1 — Geometry recovery from known relational states. [COMPLETED] Two exact quantum simulations validate the full $\Pi_{\text{loc}} \to \Pi_{\text{geom}} \to \Pi_{\text{time}}$ pipeline.

- **T1a — 1D Chain ($N = 8$ qubits).** MDS flawlessly recovered the **monotonic 1D spatial ordering** of the chain, natively selecting $D^* = 1$.
- **T1b — 2D Grid ($N = 9$ qubits, 3×3).** The pipeline successfully recovered the **exact 2D grid topology**, dynamically selecting emergent dimension $D^* = 2$.

T2 — Stability under phase-flow (locality from leakage minimization). [COMPLETED] An exhaustive search evaluated **all 105 possible 4-cell coarse-graining partitions**. The $\mathcal{L}_{\text{leak}}$ minimization **independently and uniquely recovered the contiguous 1D local blocks**. Locality was derived entirely from quantum correlations.

T4 — Emergent Gravity Well and Redshift. [COMPLETED] A localized **entropy-deficit source** ($\delta\rho_i < 0$) was injected into a 2D MI-weighted correlation graph. The resulting clock-rate potential Φ exhibited monotonic radial falloff and exact discrete gravitational redshift.

T5 — Phenomenological Stability and Radiative Cooling. [COMPLETED] CA models demonstrated **emergent spatial clustering** ("purity shields"). Disabling the cooling channel caused **total population extinction**, confirming that emergent geometry requires an effectively open thermodynamic subsystem mechanism.

C. Distinctive empirical commitments and lethal falsifiers

F1 — Dimensional Collapse. Falsified if states globally minimize complexity-stress at $D^* \to \infty$ or $D^* = 1$ when they shouldn't.

F3 — Lorentz-Violation Bound Breach. The framework predicts residual violations of order $(\ell_*/L)^n$ at the extreme UV scale. Falsified if predicted violations exceed the 10^{-14} experimental bound.

XIII. DISCUSSION AND OPEN PROBLEMS

Open problems include:

1. **Toy-model realizations of Π.** Scaling the tensor networks and GPUs to $D^* \approx 3$.
2. **GR matching as an emergent closure.**
3. **Dynamic cell net / dynamic spacetime ("geometrogenesis").**
4. **Algorithmic Isomorphism to Unsupervised Physical AI.** While this framework is constructed as a physical theory of pre-geometry, its mathematical primitives exhibit a strict structural isomorphism to the functional requirements of unsupervised representation learning and emergent physical artificial intelligence (EPAI). Standard artificial neural networks rely on rigid, human-engineered topologies and biologically implausible global backpropagation. In contrast, the POPGP pipeline naturally functions as a dynamic-topology, energy-based learning architecture driven by local thermodynamic relaxation. Specifically:
   - **Dynamic Neural Architecture (D^*):** By treating the pre-geometric substrate $(\mathcal{A}, \omega)$ as a high-dimensional data environment, the complexity-stress minimization in Π_{geom} acts as an autonomous dimensionality-reduction algorithm. The network dynamically autosizes its latent space (D^*) to optimally represent the complexity of the data without human hyperparameter tuning.
   - **Decentralized Attention via Gravity (Φ):** Structural anomalies or highly ordered data features manifest as local entropy deficits ($\delta\rho < 0$). The resulting screened graph-Laplacian natively generates a deep "gravitational" potential well ($\Phi < 0$). Because local computational time scales as $d\tau \propto e^{\Phi} dS_{\text{act}}$, the network inherently allocates significantly greater processing depth to complex data features, while rapidly fast-forwarding through unstructured noise. Gravity functions computationally as a zero-overhead physical attention mechanism.
   - **Training via Thermodynamic Relaxation:** The selection of the cell net (Π_{res}) via causally-bounded gradient descent on phase-flow leakage replaces global backpropagation. The network "learns" stable invariants purely

TABLE II. A practical test matrix

Target claim/module	Observable proxy	What must hold	What would disprove it
Stability-selected locality	Leakage/drift metrics	$\mathcal{L}_{\rm leak}$ small, drift small at fixed capacity	No stable decomposition exists without collapsing capacity
Emergent 3D space	Dimensionality	Best-fit embedding dimension ≈ 3 across scales	No stable low-dimensional embedding; dimension drifts wildly
GR closure in weak field	Extracted Φ, redshift, lensing	Matches standard weak-field phenomenology	Systematic mismatch not removable by universal calibration
Finite distinguishability	Scaling of $S(R)$ with boundary cut	$S(R)$ bounded by area scaling (operationally)	Demonstrated generic violation of area scaling at fixed regime
Junction module (γ)	Marginal dependence tests	No measurable marginal dependence in known regimes	Robust marginal dependence without classical side-channel explanation

by shedding interaction entropy into the UV buffer zone, aligning with principles of active inference and thermodynamic computing hardware.

- **Zero-Latency Consensus via Pre-Geometric Junctions:** The framework's open module for non-local "junction accessibility" (Module J) translates computationally into a mechanism for simultaneous distributed decision-making. While the emergent spatial graph ($\Pi_{\rm loc}$) imposes strict causal communication bounds (network latency), junction points act as non-local topological shortcuts in the underlying substrate algebra. In a multi-agent or distributed physical AI architecture (e.g., swarm robotics or partitioned neuromorphic clusters), these junctions permit geometrically separated nodes to execute perfectly synchronous global state updates and simultaneous consensus, entirely bypassing the geometric latency bottlenecks of the emergent manifold.

The application of this phase-flow relaxation as a native algorithm for thermodynamic or neuromorphic computation is a subject of separate, forthcoming investigation.

XIV. WORKED EXAMPLES (OPERATIONAL EXTRACTION OF STANDARD OBSERVABLES)

A. Slicing, gauge, and the Shift Vector via Optimal Transport

1. *The Shift Vector from Intrinsic Optimal Transport*

$$v^a_{ij} \approx \mathrm{Log}_{x_i}(x_j) \tag{41}$$

$$N^a(x_i)\Delta s := \sum_j \frac{\gamma^*_{ij}}{\mu_i} v^a_{ij} \tag{42}$$

2. *Discrete Weak-field diagnostic potentials*

Time potential Φ_t at node i: $\Phi_t(i) := \Phi_i - \Phi_B$. The discrete gravitational redshift:

$$1 + z = \exp(\Phi_A - \Phi_B) \tag{43}$$

Space potential Φ_s from discrete conformal factors:

$$a_s(i) := \left(\frac{\det(h_{ab}(i))}{\det(h_B)}\right)^{1/6} \tag{44}$$

B. Worked example A — isolated spherical mass (weak-field, static)

1. *Discrete Geodesic Shooting*

1. **Null Geodesics:**

$$L_{\rm opt} = \sum_{\text{edge}=(i,j)} \frac{\ell_{ij}}{\beta_{\rm avg}(i,j)} \tag{45}$$

2. **Redshift:**

$$1 + z = \frac{\beta(A)}{\beta(B)} = \frac{\exp(\Phi_A)}{\exp(\Phi_B)} \tag{46}$$

3. **Shapiro Delay:**

$$\Delta\tau \approx \sum_{\text{edges}} \ell_{ij}\big(1 - \exp(\Phi_{\rm avg})\big) \tag{47}$$

C. Worked example C — homogeneous cosmology patch (FLRW-like)

1. *Discrete Scale Factor*

$$V_P(t_k) := \sum_{T\in\mathcal{T}_{\mathrm{VR}}} \mathrm{Vol}(T) \tag{48}$$

$$a(t_k) := \left(\frac{V_P(t_k)}{V_P(t_0)}\right)^{1/3} \tag{49}$$

2. *Hubble Rate*

$$H(t_k) \approx \frac{1}{a(t_k)} \frac{a(t_{k+1}) - a(t_k)}{\Delta\bar{\tau}_k} \tag{50}$$

3. *Failure Modes (Falsification)*

- **Volume Collapse:** If the Vietoris-Rips complex degenerates (slivers, zero-volume tetrahedra) despite a growing V_P, the geometry is not manifold-like.
- **Anisotropy:** If $a(t)$ differs significantly when measured along different graph axes (using discrete direction-dependent correlators), the emergent space is not FLRW.

D. How these worked examples connect to falsification

These examples define *what to compute* from a candidate instantiation $(\mathcal{A}, \omega)$ and the specified selection rule E. The framework is disfavored if, under its own constraints (symmetry commutation, retention, area capacity, and stability), it cannot realize:

- a weak-field isolated-mass regime reproducing redshift and lensing;
- multi-lens weak-field behavior with approximate superposition and shear;
- a homogeneous expanding patch with robust scale-factor diagnostics.

In other words: these examples translate “emergent geometry” from a slogan into a set of explicit computational tests.